arm 2 and received reward 9. This list would grow longer as we played the game.

There's a much simpler approach, however, since we really only need to keep track of the average reward for each arm—we don't need to store each observation. Recall that to calculate the mean of a list of numbers, x_i (indexed by i), we simply need sum up all the x_i values and then divide by the number of x_i, which we will denote k. The mean is often denoted with the Greek letter μ (mu).

$$\mu = \frac{1}{k}\sum_{i} x_i$$

The Greek uppercase symbol Σ (sigma) is used to denote a summation operation. The i notation underneath means we sum each element, x_i. It's basically the math equivalent of a `for` loop like:

```
sum = 0
x = [4,5,6,7]
for j in range(len(x)):
    sum = sum + x[j]
```

If we already have an average reward μ for a particular arm, we can update this average when we get a new reward by recomputing the average. We basically need to undo the average and then recompute it. To undo it, we multiply μ by the total number of values, k. Of course, this just gives us the sum, not the original set of values—you can't undo a sum. But the total number is what we need to recompute the average with a new value. We just add this sum to the new value and divide by $k + 1$, the new total number of values.

$$\mu_{new} = \frac{k \cdot \mu_{old} + x}{k + 1}$$

We can use this equation to continually update the average reward observed for each arm as we collect new data, and this way we only need to keep track of two numbers for each arm: k, the number of values observed, and